

Soheil Sayah Varg

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Education

Sharif University of Technology

B.Sc. in Computer Engineering

Tehran, Iran

Sep 2023 – Jul 2027

- **Cumulative GPA: 19.45 / 20.0**, ranked **15th** of 185 students. Departmental average for this cohort: **17.45**. University-wide: 16.11.
- Awarded 7 credits in physics and admitted without entrance exam, both under the National Olympiad medalist provision.

AE High School

Diploma, Mathematics & Physics

Tehran, Iran

Sep 2020 – Jun 2023

- GPA: 19.25 / 20.0

Honors and Awards

2023: Gold Medal, Iranian National Physics Olympiad — ranked **7th** nationwide among more than 10,000 participants, and **2nd** nationwide in the experimental (laboratory) examination.

Research Interests

Machine Learning Theory • Probabilistic Inference and Sampling • Bandits and Reinforcement Learning • Large Language Models

Research Experience

Data Science & Machine Learning Laboratory (DML), Sharif University of Technology

Exact Controlled Generation from Tilted Autoregressive Targets

Dec 2025 – Present

- Supervised by **Dr. Ali Rostami**. Led the research and algorithm design for sampling from reward-tilted autoregressive language models using a particle-Gibbs ladder over a batched proposal chain.
- Worked on correctness and target-invariance arguments, implemented the method, and ran its experiments.
- **First-author manuscript in preparation.**

Graduate Deep RL project, team of 3, Sharif University of Technology

Model-Based Reinforcement Learning in Confounded POMDPs

Feb 2026 – Aug 2026

- Led a three-person project implementing and evaluating the method of Hong, Qi, and Xu (ICML 2024). Built the experimental pipeline and implemented bridge-function identification using negative-control proxies.
- Validated it against an exact oracle and designed experiments to examine its finite-sample behaviour. Took responsibility for writing the final results.

May 2026 – Present: Bandit Literature Review, supervised by Dr. Amir Najafi — studying batched bandits, privacy-constrained linear bandits, and misspecified kernelized bandit optimization. Derived a regret analysis for two-arm explore-then-commit.

Selected Projects

Game Theory: War of Attrition under Uncertainty — Python, PyTorch

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- Bayesian type filtering and potential-based reward shaping for a multi-stage war of attrition with incomplete information; behavioural features and self-play evaluation.

Information Retrieval: Modern Information Retrieval System — Python, PyTorch, FAISS

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- SPLADE + dense retrieval fused by reciprocal rank fusion, cross-encoder reranking, QLoRA fine-tuning for generation.

Deep RL: Segmentation and Deep RL — Python, PyTorch

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- Attention U-Net variants for road segmentation and Soft Actor-Critic for continuous control, built from scratch in PyTorch.

Optimization: Convex Optimization Algorithms — Python, NumPy, CVXPY

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- Chambolle-Pock and ADMM implemented from their update rules; ADMM matches CVXPY to 0.02 dB PSNR in 0.2 s.

Applied Probability: Market Predictability Study — Python, statsmodels

🌐 repo

- Showed the claimed signal was a multiple-comparisons artifact; a coin flip reproduced its Sharpe ratio.

Systems: **NeoGit** — a version control system in C

 **repo**

○ Git-like CLI on raw POSIX file I/O: staging, commits, branching, merge, stashing, pre-commit hooks.

Teaching Experience

Sharif University of Technology

Teaching Assistant

2024 – Present

- Multi-Agent Reinforcement Learning (1 time); Machine Learning (2 times); Deep Learning (1 time); High-Dimensional Statistics (1 time).
- Stochastic Processes (2 times); Probability & Statistics (3 times); Artificial Intelligence (1 time); Data Science (1 time).
- Signals & Systems (1 time, **Head TA**); Linear Algebra (2 times); Data Structures & Algorithms (1 time); Logic Design (1 time); Fundamentals of Programming (1 time).

18 course-semester appointments, including current Fall 2026 appointments. Have begun preparing theoretical questions for the upcoming term. As Head TA, coordinated the teaching staff for a 60-student course.

Aug 2023 – Present: Co-founder and physics instructor, Resonance (resonanceoly.ir) — an online Physics Olympiad school, started as a small study group, reaching students who have no selective high school near them. Teach the mechanics, electromagnetics and laboratory tracks, and write problem sets and mock exams for the national selection rounds.

2022 – Present: Physics Olympiad instructor, theory and **experimental / laboratory** tracks — core olympiad staff at **Allameh Helli 3** (NODET, Iran's national programme for exceptional talents) and instructor at AE High School. Also introductory Python at AE Middle School and lower-secondary physics at Mahdavi School.

Leadership and Activities

2025 – 2026: Peer Mentor, First-Year Peer Mentorship Program (Hamyar), Sharif University of Technology — served in Dr. Kasaei's mentoring team as a liaison and point of guidance for assigned incoming students during their transition to university.

May 2025: Juror, 13th Persian Young Naturalists' Tournament (PYNT), Tehran — an IYPT-format research tournament run by the Ariaian Young Innovative Minds Institute under the World Federation of Physics Competitions. Judged team defenses of open-ended physics research problems.

2024 – 2025: Deputy Chairman, CEI 2024 (Computer Engineering Introduction event) • **Head of Purchase**, WSS 11 (Winter Seminar Series) • Sponsorship Staff, WSS 10.

2023 – 2025: Head of Purchase & Scientific Staff, Bugsbuzzy game development contest • Executive Staff, ICPC 2023 & 2025 • Rayan AI 2024.

Selected Coursework ([†] graduate-level)

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| ○ Stochastic Processes [†] (20.0) | ○ Computer Simulation (20.0) |
| ○ Machine Learning (20.0) | ○ Numerical Computation (19.6) |
| ○ Artificial Intelligence (20.0) | ○ Data Structures & Algorithms (20.0) |
| ○ Convex Optimization I (19.5) | ○ Signals & Systems (20.0) |
| ○ Linear Algebra (20.0) | ○ Reinforcement Learning [†] (20.0) |
| ○ Probability & Statistics (20.0) | ○ Game Theory (20.0) |

Additional: Modern Information Retrieval (19.8), Design of Algorithms (20.0), Database Design (20.0).

Planned: Machine Learning Theory[†] with Dr. Amir Najafi.

Technical Skills

Languages: Python (proficient), C, C++, Java, SQL, MATLAB, $\LaTeX$

ML & Numerics: PyTorch, NumPy, SciPy, scikit-learn, statsmodels, pandas, Matplotlib, Hugging Face Transformers

Optimization & IR: CVXPY, FAISS, Gymnasium

Tools: Git, Linux, Docker, PostgreSQL, shell scripting, Proteus